

NURTURE COURSE
SOLUTIONS OF TRIANGLE

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SOLUTIONS OF TRIANGLE

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JEE (ADVANCED) SYLLABUS :

Solutions of Triangle : Relations between sides and angles of a triangle, sine rule, cosine rule, half-angle formula and the area of a triangle.

SOLUTIONS OF TRIANGLE

The process of calculating the sides and angles of triangle using given information is called solution of triangle.

In a $\triangle ABC$, the angles are denoted by capital letters A, B and C and the length of the sides opposite these angle are denoted by small letter a, b and c respectively.

1. SINE FORMULAE :

In any triangle ABC

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = \lambda = \frac{abc}{2\Delta} = 2R$$

where R is circumradius and Δ is area of triangle.

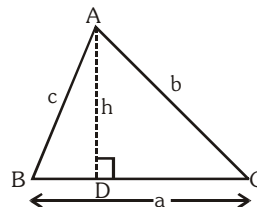


Illustration 1 : Angles of a triangle are in 4 : 1 : 1 ratio. The ratio between its greatest side and perimeter is

- (A) $\frac{3}{2+\sqrt{3}}$ (B) $\frac{\sqrt{3}}{2+\sqrt{3}}$ (C) $\frac{\sqrt{3}}{2-\sqrt{3}}$ (D) $\frac{1}{2+\sqrt{3}}$

Solution : Angles are in ratio 4 : 1 : 1.

$\Rightarrow$ angles are $120^\circ, 30^\circ, 30^\circ$.

If sides opposite to these angles are a, b, c respectively, then a will be the greatest side.

Now from sine formula $\frac{a}{\sin 120^\circ} = \frac{b}{\sin 30^\circ} = \frac{c}{\sin 30^\circ}$

$$\Rightarrow \frac{a}{\sqrt{3}/2} = \frac{b}{1/2} = \frac{c}{1/2}$$

$$\Rightarrow \frac{a}{\sqrt{3}} = \frac{b}{1} = \frac{c}{1} = k \text{ (say)}$$

then $a = \sqrt{3}k$, perimeter = $(2 + \sqrt{3})k$

$$\therefore \text{required ratio} = \frac{\sqrt{3}k}{(2 + \sqrt{3})k} = \frac{\sqrt{3}}{2 + \sqrt{3}}$$

Ans. (B)

Illustration 2 : In triangle ABC, if $b = 3$, $c = 4$ and $\angle B = \pi/3$, then number of such triangles is -

- (A) 1 (B) 2 (C) 0 (D) infinite

Solution : Using sine formulae $\frac{\sin B}{b} = \frac{\sin C}{c}$

$$\Rightarrow \frac{\sin \pi/3}{3} = \frac{\sin C}{4} \Rightarrow \frac{\sqrt{3}}{6} = \frac{\sin C}{4} \Rightarrow \sin C = \frac{2}{\sqrt{3}} > 1 \text{ which is not possible.}$$

Hence there exist no triangle with given elements.

Ans. (C)

Illustration 3 : The sides of a triangle are three consecutive natural numbers and its largest angle is twice the smallest one. Determine the sides of the triangle.

Solution : Let the sides be $n, n+1, n+2$ cms.

i.e. $AC = n, AB = n+1, BC = n+2$

Smallest angle is B and largest one is A.

Here, $\angle A = 2\angle B$

Also, $\angle A + \angle B + \angle C = 180^\circ$

$$\Rightarrow 3\angle B + \angle C = 180^\circ \Rightarrow \angle C = 180^\circ - 3\angle B$$

We have, sine law as,

$$\frac{\sin A}{n+2} = \frac{\sin B}{n} = \frac{\sin C}{n+1} \Rightarrow \frac{\sin 2B}{n+2} = \frac{\sin B}{n} = \frac{\sin(180-3B)}{n+1}$$

$$\Rightarrow \frac{\sin 2B}{n+2} = \frac{\sin B}{n} = \frac{\sin 3B}{n+1}$$

(i) (ii) (iii)

from (i) and (ii);

$$\frac{2 \sin B \cos B}{n+2} = \frac{\sin B}{n} \Rightarrow \cos B = \frac{n+2}{2n} \quad \dots\dots (iv)$$

and from (ii) and (iii);

$$\frac{\sin B}{n} = \frac{3 \sin B - 4 \sin^3 B}{n+1} \Rightarrow \frac{\sin B}{n} = \frac{\sin B(3 - 4 \sin^2 B)}{n+1}$$

$$\Rightarrow \frac{n+1}{n} = 3 - 4(1 - \cos^2 B) \quad \dots\dots (v)$$

from (iv) and (v), we get

$$\frac{n+1}{n} = -1 + 4 \left(\frac{n+2}{2n} \right)^2 \Rightarrow \frac{n+1}{n} + 1 = \left(\frac{n^2 + 4n + 4}{n^2} \right)$$

$$\Rightarrow \frac{2n+1}{n} = \frac{n^2 + 4n + 4}{n^2} \Rightarrow 2n^2 + n = n^2 + 4n + 4$$

$$\Rightarrow n^2 - 3n - 4 = 0 \Rightarrow (n-4)(n+1) = 0$$

$$n = 4 \text{ or } -1$$

where $n \neq -1$

$\therefore n = 4$. Hence the sides are 4, 5, 6

Ans.

Do yourself - 1 :

- (i) If in a $\triangle ABC$, $\angle A = \frac{\pi}{6}$ and $b : c = 2 : \sqrt{3}$, find $\angle B$.
- (ii) Show that, in any $\triangle ABC$: $a \sin(B-C) + b \sin(C-A) + c \sin(A-B) = 0$.
- (iii) If in a $\triangle ABC$, $\frac{\sin A}{\sin C} = \frac{\sin(A-B)}{\sin(B-C)}$, show that a^2, b^2, c^2 are in A.P.

2. COSINE FORMULAE :

$$(a) \quad \cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\text{(b)} \quad \cos B = \frac{c^2 + a^2 - b^2}{2ca}$$

$$\text{(c) } \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\text{or } a^2 = b^2 + c^2 - 2bc \cos A$$

Illustration 4 : In a triangle ABC, if $B = 30^\circ$ and $c = \sqrt{3} b$, then A can be equal to -

- (A) 45° (B) 60° (C) 90° (D) 120°

Solution : We have $\cos B = \frac{c^2 + a^2 - b^2}{2ca} \Rightarrow \frac{\sqrt{3}}{2} = \frac{3b^2 + a^2 - b^2}{2 \times \sqrt{3}b \times a}$

$$\Rightarrow a^2 - 3ab + 2b^2 = 0 \Rightarrow (a - 2b)(a - b) = 0$$

$$\Rightarrow \text{Either } a = b \Rightarrow A = 30^\circ$$

$$\text{or } a = 2b \Rightarrow a^2 = 4b^2 = b^2 + c^2 \Rightarrow A = 90^\circ.$$

Ans. (C)

Illustration 5 : In a triangle ABC, $(a^2 - b^2 - c^2) \tan A + (a^2 - b^2 + c^2) \tan B$ is equal to -

- (A) $(a^2 + b^2 - c^2) \tan C$ (B) $(a^2 + b^2 + c^2) \tan C$
(C) $(b^2 + c^2 - a^2) \tan C$ (D) none of these

Solution : Using cosine law :

The given expression is equal to $-2bc \cos A \tan A + 2ac \cos B \tan B$

$$= 2abc \left(-\frac{\sin A}{a} + \frac{\sin B}{b} \right) = 0 \quad \text{Ans. (D)}$$

Do yourself - 2 :

- (i) If $a : b : c = 4 : 5 : 6$, then show that $\angle C = 2\angle A$.
- (ii) In any $\triangle ABC$, prove that

$$(a) \quad \frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c} = \frac{a^2 + b^2 + c^2}{2abc}$$

$$(b) \quad \frac{b^2}{a} \cos A + \frac{c^2}{b} \cos B + \frac{a^2}{c} \cos C = \frac{a^4 + b^4 + c^4}{2abc}$$

3. PROJECTION FORMULAE :

- (a) $b \cos C + c \cos B = a$ (b) $c \cos A + a \cos C = b$ (c) $a \cos B + b \cos A = c$

Illustration 6 : In a $\triangle ABC$, $c \cos^2 \frac{A}{2} + a \cos^2 \frac{C}{2} = \frac{3b}{2}$, then show a, b, c are in A.P.

Solution : Here, $\frac{c}{2}(1 + \cos A) + \frac{a}{2}(1 + \cos C) = \frac{3b}{2}$
 $\Rightarrow a + c + (c \cos A + a \cos C) = 3b$
 $\Rightarrow a + c + b = 3b$ {using projection formula}
 $\Rightarrow a + c = 2b$
 which shows a, b, c are in A.P.

Do yourself - 3 :

(i) In a $\triangle ABC$, if $\angle A = \frac{\pi}{4}$, $\angle B = \frac{5\pi}{12}$, show that $a + c\sqrt{2} = 2b$.

(ii) In a $\triangle ABC$, prove that : (a) $b(a \cos C - c \cos A) = a^2 - c^2$ (b) $2 \left(b \cos^2 \frac{C}{2} + c \cos^2 \frac{B}{2} \right) = a + b + c$

4. NAPIER'S ANALOGY (TANGENT RULE) :

$$(a) \tan\left(\frac{B-C}{2}\right) = \frac{b-c}{b+c} \cot \frac{A}{2} \quad (b) \tan\left(\frac{C-A}{2}\right) = \frac{c-a}{c+a} \cot \frac{B}{2} \quad (c) \tan\left(\frac{A-B}{2}\right) = \frac{a-b}{a+b} \cot \frac{C}{2}$$

Illustration 7 : In a $\triangle ABC$, the tangent of half the difference of two angles is one-third the tangent of half the sum of the angles. Determine the ratio of the sides opposite to the angles.

Solution : Here, $\tan\left(\frac{A-B}{2}\right) = \frac{1}{3} \tan\left(\frac{A+B}{2}\right)$ (i)

using Napier's analogy, $\tan\left(\frac{A-B}{2}\right) = \frac{a-b}{a+b} \cot\left(\frac{C}{2}\right)$ (ii)

from (i) & (ii) ;

$$\frac{1}{3} \tan\left(\frac{A+B}{2}\right) = \frac{a-b}{a+b} \cot\left(\frac{C}{2}\right) \Rightarrow \frac{1}{3} \cot\left(\frac{C}{2}\right) = \frac{a-b}{a+b} \cot\left(\frac{C}{2}\right)$$

$$\{ \text{as } A + B + C = \pi \therefore \tan\left(\frac{B+C}{2}\right) = \tan\left(\frac{\pi}{2} - \frac{C}{2}\right) = \cot \frac{C}{2} \}$$

$$\Rightarrow \frac{a-b}{a+b} = \frac{1}{3} \quad \text{or} \quad 3a - 3b = a + b$$

$$2a = 4b \quad \text{or} \quad \frac{a}{b} = \frac{2}{1} \Rightarrow \frac{b}{a} = \frac{1}{2}$$

Thus the ratio of the sides opposite to the angles is $b : a = 1 : 2$.

Ans.

Do yourself - 4 :

(i) In any $\triangle ABC$, prove that $\frac{b-c}{b+c} = \frac{\tan\left(\frac{B-C}{2}\right)}{\tan\left(\frac{B+C}{2}\right)}$

(ii) If $\triangle ABC$ is right angled at C, prove that : (a) $\tan \frac{A}{2} = \sqrt{\frac{c-b}{c+b}}$ (b) $\sin(A-B) = \frac{a^2 - b^2}{a^2 + b^2}$

5. HALF ANGLE FORMULAE :

$s = \frac{a+b+c}{2}$ = semi-perimeter of triangle.

(a) (i) $\sin \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}$ (ii) $\sin \frac{B}{2} = \sqrt{\frac{(s-c)(s-a)}{ca}}$ (iii) $\sin \frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{ab}}$

(b) (i) $\cos \frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}$ (ii) $\cos \frac{B}{2} = \sqrt{\frac{s(s-b)}{ca}}$ (iii) $\cos \frac{C}{2} = \sqrt{\frac{s(s-c)}{ab}}$

(c) (i) $\tan \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$ (ii) $\tan \frac{B}{2} = \sqrt{\frac{(s-c)(s-a)}{s(s-b)}}$ (iii) $\tan \frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{s(s-c)}}$
 $= \frac{\Delta}{s(s-a)}$ $= \frac{\Delta}{s(s-b)}$ $= \frac{\Delta}{s(s-c)}$

(d) Area of Triangle

$\Delta = \sqrt{s(s-a)(s-b)(s-c)} = \frac{1}{2}bc \sin A = \frac{1}{2}ca \sin B = \frac{1}{2}ab \sin C = \frac{1}{2}ap_1 = \frac{1}{2}bp_2 = \frac{1}{2}cp_3$,
 where p_1, p_2, p_3 are altitudes from vertices A, B, C respectively.

Illustration 8 : If in a triangle ABC, CD is the angle bisector of the angle ACB, then CD is equal to-

(A) $\frac{a+b}{2ab} \cos \frac{C}{2}$ (B) $\frac{2ab}{a+b} \sin \frac{C}{2}$ (C) $\frac{2ab}{a+b} \cos \frac{C}{2}$ (D) $\frac{b \sin \angle DAC}{\sin(B+C/2)}$

Solution :

$\triangle CAB = \triangle CAD + \triangle CDB$

$\Rightarrow \frac{1}{2} ab \sin C = \frac{1}{2} b \cdot CD \cdot \sin\left(\frac{C}{2}\right) + \frac{1}{2} a \cdot CD \cdot \sin\left(\frac{C}{2}\right)$

$\Rightarrow CD(a+b) \sin\left(\frac{C}{2}\right) = ab \left(2 \sin\left(\frac{C}{2}\right) \cos\left(\frac{C}{2}\right)\right)$

So $CD = \frac{2ab \cos(C/2)}{(a+b)}$

and in $\triangle CAD$, $\frac{CD}{\sin \angle DAC} = \frac{b}{\sin \angle CDA}$ (by sine rule)

$\Rightarrow CD = \frac{b \sin \angle DAC}{\sin(B+C/2)}$

Ans. (C,D)

Illustration 9 : If Δ is the area and $2s$ the sum of the sides of a triangle, then show $\Delta \leq \frac{s^2}{3\sqrt{3}}$.

Solution : We have, $2s = a + b + c$, $\Delta^2 = s(s-a)(s-b)(s-c)$

Now, A.M. $\geq$ G.M.

$$\frac{(s-a) + (s-b) + (s-c)}{3} \geq \{(s-a)(s-b)(s-c)\}^{1/3}$$

$$\text{or } \frac{3s-2s}{3} \geq \left(\frac{\Delta^2}{s}\right)^{1/3}$$

$$\text{or } \frac{s}{3} \geq \left(\frac{\Delta^2}{s}\right)^{1/3}$$

$$\text{or } \frac{\Delta^2}{s} \leq \frac{s^3}{27} \Rightarrow \Delta \leq \frac{s^2}{3\sqrt{3}}$$

Ans.

Do yourself - 5 :

(i) Given $a = 6$, $b = 8$, $c = 10$. Find

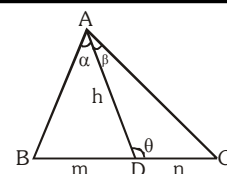
(a) $\sin A$ (b) $\tan A$ (c) $\sin \frac{A}{2}$ (d) $\cos \frac{A}{2}$ (e) $\tan \frac{A}{2}$ (f) Δ

(ii) Prove that in any ΔABC , $(abc) \sin \frac{A}{2} \cdot \sin \frac{B}{2} \cdot \sin \frac{C}{2} = \Delta^2$.

6. m-n THEOREM :

$$(m+n) \cot \theta = m \cot \alpha + n \cot \beta$$

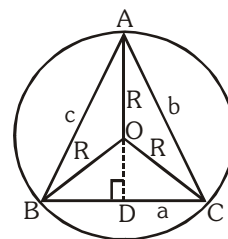
$$(m+n) \cot \theta = n \cot B + m \cot C.$$



7. RADIUS OF THE CIRCUMCIRCLE 'R' :

Circumcentre is the point of intersection of perpendicular bisectors of the sides and distance between circumcentre & vertex of triangle is called circumradius 'R'.

$$R = \frac{a}{2 \sin A} = \frac{b}{2 \sin B} = \frac{c}{2 \sin C} = \frac{abc}{4\Delta}.$$



8. RADIUS OF THE INCIRCLE 'r' :

Point of intersection of internal angle bisectors is incentre and perpendicular distance of incentre from any side is called inradius 'r'.

$$r = \frac{\Delta}{s} = (s-a) \tan \frac{A}{2} = (s-b) \tan \frac{B}{2} = (s-c) \tan \frac{C}{2} = 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}.$$

$$= a \frac{\sin \frac{B}{2} \sin \frac{C}{2}}{\cos \frac{A}{2}} = b \frac{\sin \frac{A}{2} \sin \frac{C}{2}}{\cos \frac{B}{2}} = c \frac{\sin \frac{B}{2} \sin \frac{A}{2}}{\cos \frac{C}{2}}$$

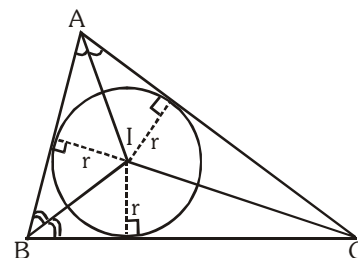


Illustration 10 : In a triangle ABC, if $a : b : c = 4 : 5 : 6$, then ratio between its circumradius and inradius is-

- (A) $\frac{16}{7}$ (B) $\frac{16}{9}$ (C) $\frac{7}{16}$ (D) $\frac{11}{7}$

Solution : $\frac{R}{r} = \frac{abc}{4\Delta} \cdot \frac{\Delta}{s} = \frac{(abc)s}{4\Delta^2} \Rightarrow \frac{R}{r} = \frac{abc}{4(s-a)(s-b)(s-c)} \dots(i)$

$\therefore a : b : c = 4 : 5 : 6 \Rightarrow \frac{a}{4} = \frac{b}{5} = \frac{c}{6} = k \text{ (say)}$

$\Rightarrow a = 4k, b = 5k, c = 6k$

$\therefore s = \frac{a+b+c}{2} = \frac{15k}{2}, s-a = \frac{7k}{2}, s-b = \frac{5k}{2}, s-c = \frac{3k}{2}$

using (i) in these values $\frac{R}{r} = \frac{(4k)(5k)(6k)}{4 \left(\frac{7k}{2}\right) \left(\frac{5k}{2}\right) \left(\frac{3k}{2}\right)} = \frac{16}{7}$ **Ans. (A)**

Illustration 11 : If A, B, C are the angles of a triangle, prove that : $\cos A + \cos B + \cos C = 1 + \frac{r}{R}$.

Solution : $\cos A + \cos B + \cos C = 2 \cos\left(\frac{A+B}{2}\right) \cdot \cos\left(\frac{A-B}{2}\right) + \cos C$

$= 2 \sin \frac{C}{2} \cdot \cos\left(\frac{A-B}{2}\right) + 1 - 2 \sin^2 \frac{C}{2} = 1 + 2 \sin \frac{C}{2} \left[\cos\left(\frac{A-B}{2}\right) - \sin\left(\frac{C}{2}\right) \right]$

$= 1 + 2 \sin \frac{C}{2} \left[\cos\left(\frac{A-B}{2}\right) - \cos\left(\frac{A+B}{2}\right) \right] \quad \left\{ \because \frac{C}{2} = 90^\circ - \left(\frac{A+B}{2}\right) \right\}$

$= 1 + 2 \sin \frac{C}{2} \cdot 2 \sin \frac{A}{2} \cdot \sin \frac{B}{2} = 1 + 4 \sin \frac{A}{2} \cdot \sin \frac{B}{2} \cdot \sin \frac{C}{2}$

$= 1 + \frac{r}{R} \quad \{ \text{as, } r = 4R \sin \frac{A}{2} \cdot \sin \frac{B}{2} \cdot \sin \frac{C}{2} \}$

$\Rightarrow \cos A + \cos B + \cos C = 1 + \frac{r}{R}$. Hence proved.

Do yourself - 6 :

(i) If in $\triangle ABC$, $a = 3$, $b = 4$ and $c = 5$, find

- (a) Δ (b) R (c) r

(ii) In a $\triangle ABC$, show that :

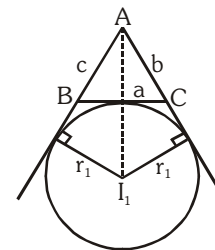
(a) $\frac{a^2 - b^2}{c} = 2R \sin(A-B)$ (b) $r \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} = \frac{\Delta}{4R}$ (c) $a + b + c = \frac{abc}{2Rr}$

(iii) Let Δ & Δ' denote the areas of a Δ and that of its incircle. Prove that

$\Delta : \Delta' = \left(\cot \frac{A}{2} \cdot \cot \frac{B}{2} \cdot \cot \frac{C}{2} \right) : \pi$

9. RADII OF THE EX-CIRCLES :

Point of intersection of two external angles and one internal angle bisectors is excentre and perpendicular distance of excentre from any side is called exradius. If r_1 is the radius of escribed circle opposite to $\angle A$ of $\triangle ABC$ and so on, then -



$$(a) \quad r_1 = \frac{\Delta}{s-a} = s \tan \frac{A}{2} = 4R \sin \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} = \frac{a \cos \frac{B}{2} \cos \frac{C}{2}}{\cos \frac{A}{2}}$$

$$(b) \quad r_2 = \frac{\Delta}{s-b} = s \tan \frac{B}{2} = 4R \cos \frac{A}{2} \sin \frac{B}{2} \cos \frac{C}{2} = \frac{b \cos \frac{A}{2} \cos \frac{C}{2}}{\cos \frac{B}{2}}$$

$$(c) \quad r_3 = \frac{\Delta}{s-c} = s \tan \frac{C}{2} = 4R \cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2} = \frac{c \cos \frac{A}{2} \cos \frac{B}{2}}{\cos \frac{C}{2}}$$

I_1, I_2 and I_3 are taken as ex-centre opposite to vertex A, B, C respectively.

Illustration 12 : Value of the expression $\frac{b-c}{r_1} + \frac{c-a}{r_2} + \frac{a-b}{r_3}$ is equal to -

- (A) 1 (B) 2 (C) 3 (D) 0

Solution :

$$\frac{(b-c)}{r_1} + \frac{(c-a)}{r_2} + \frac{(a-b)}{r_3}$$

$$\Rightarrow (b-c) \left(\frac{s-a}{\Delta} \right) + (c-a) \left(\frac{s-b}{\Delta} \right) + (a-b) \left(\frac{s-c}{\Delta} \right)$$

$$\Rightarrow \frac{(s-a)(b-c) + (s-b)(c-a) + (s-c)(a-b)}{\Delta}$$

$$= \frac{s(b-c+c-a+a-b) - [ab-ac+bc-ba+ac-bc]}{\Delta} = \frac{0}{\Delta} = 0$$

$$\text{Thus, } \frac{b-c}{r_1} + \frac{c-a}{r_2} + \frac{a-b}{r_3} = 0$$

Ans. (D)

Illustration 13 : If $r_1 = r_2 + r_3 + r$, prove that the triangle is right angled.

Solution :

We have, $r_1 - r = r_2 + r_3$

$$\Rightarrow \frac{\Delta}{s-a} - \frac{\Delta}{s} = \frac{\Delta}{s-b} + \frac{\Delta}{s-c}$$

$$\Rightarrow \frac{a}{s(s-a)} = \frac{2s-(b+c)}{(s-b)(s-c)}$$

$$\Rightarrow \frac{a}{s(s-a)} = \frac{a}{(s-b)(s-c)}$$

$$\Rightarrow \frac{s-s+a}{s(s-a)} = \frac{s-c+s-b}{(s-b)(s-c)}$$

$$\{as, 2s = a + b + c\}$$

$$\Rightarrow s^2 - (b+c)s + bc = s^2 - as$$

$$\begin{aligned} \Rightarrow s(-a + b + c) &= bc & \Rightarrow \frac{(b+c-a)(a+b+c)}{2} &= bc \\ \Rightarrow (b+c)^2 - (a)^2 &= 2bc & \Rightarrow b^2 + c^2 + 2bc - a^2 &= 2bc \\ \Rightarrow b^2 + c^2 &= a^2 \\ \therefore \angle A &= 90^\circ. \end{aligned}$$

Ans.

Do yourself - 7 :

(i) In an equilateral $\triangle ABC$, $R = 2$, find

(a) r (b) r_1 (c) a

(ii) In a $\triangle ABC$, show that

$$\begin{aligned} (a) \quad r_1 r_2 + r_2 r_3 + r_3 r_1 &= s^2 & (b) \quad \frac{1}{4} r^2 s^2 \left(\frac{1}{r} - \frac{1}{r_1} \right) \left(\frac{1}{r} - \frac{1}{r_2} \right) \left(\frac{1}{r} - \frac{1}{r_3} \right) &= R \\ (c) \quad \sqrt{r r_1 r_2 r_3} &= \Delta \end{aligned}$$

10. ANGLE BISECTORS & MEDIANS :

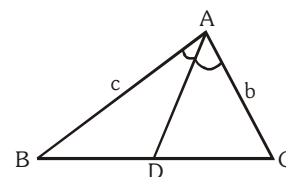
An angle bisector divides the base in the ratio of corresponding sides.

$$\frac{BD}{CD} = \frac{c}{b} \Rightarrow BD = \frac{ac}{b+c} \quad \& \quad CD = \frac{ab}{b+c}$$

If m_a and β_a are the lengths of a median and an angle bisector from the angle A then,

$$m_a = \frac{1}{2} \sqrt{2b^2 + 2c^2 - a^2} \quad \text{and} \quad \beta_a = \frac{2bc \cos \frac{A}{2}}{b+c}$$

$$\text{Note that } m_a^2 + m_b^2 + m_c^2 = \frac{3}{4}(a^2 + b^2 + c^2)$$

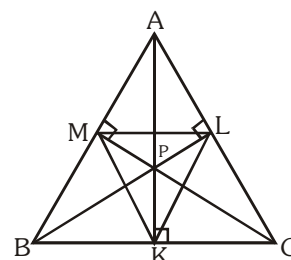


11. ORTHOCENTRE :

(a) Point of intersection of altitudes is orthocentre & the triangle KLM which is formed by joining the feet of the altitudes is called the pedal triangle.

(b) The distances of the orthocentre from the angular points of the $\triangle ABC$ are $2R \cos A$, $2R \cos B$, & $2R \cos C$.

(c) The distance of P from sides are $2R \cos B \cos C$, $2R \cos C \cos A$ and $2R \cos A \cos B$.



Do yourself - 8 :

- (i) If x, y, z are the distance of the vertices of $\triangle ABC$ respectively from the orthocentre, then

prove that $\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = \frac{abc}{xyz}$

- (ii) If p_1, p_2, p_3 are respectively the perpendiculars from the vertices of a triangle to the opposite sides, prove that

(a) $p_1 p_2 p_3 = \frac{a^2 b^2 c^2}{8R^3}$

(b) $\Delta = \sqrt{\frac{1}{2} R p_1 p_2 p_3}$

- (iii) In a $\triangle ABC$, AD is altitude and H is the orthocentre prove that $AH : DH = (\tan B + \tan C) : \tan A$

- (iv) In a $\triangle ABC$, the lengths of the bisectors of the angle A, B and C are x, y, z respectively.

Show that $\frac{1}{x} \cos \frac{A}{2} + \frac{1}{y} \cos \frac{B}{2} + \frac{1}{z} \cos \frac{C}{2} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}$.

12. THE DISTANCES BETWEEN THE SPECIAL POINTS :

- (a) The distance between circumcentre and orthocentre is $= R\sqrt{1 - 8 \cos A \cos B \cos C}$

- (b) The distance between circumcentre and incentre is $= \sqrt{R^2 - 2Rr}$

- (c) The distance between incentre and orthocentre is $= \sqrt{2r^2 - 4R^2 \cos A \cos B \cos C}$

- (d) The distances between circumcentre & excentres are

$$OI_1 = R \sqrt{1 + 8 \sin \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}} = \sqrt{R^2 + 2Rr_1} \text{ \& so on.}$$

Illustration 14 : Prove that the distance between the circumcentre and the orthocentre of a triangle ABC is $R\sqrt{1 - 8 \cos A \cos B \cos C}$.

Solution : Let O and P be the circumcentre and the orthocentre respectively. If OF is the perpendicular to AB , we have $\angle OAF = 90^\circ - \angle AOF = 90^\circ - C$. Also $\angle PAL = 90^\circ - C$.

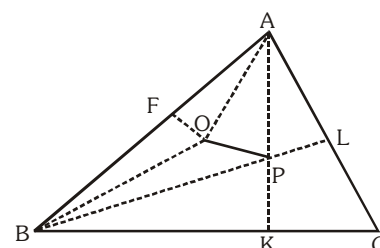
$$\text{Hence, } \angle OAP = A - \angle OAF - \angle PAL = A - 2(90^\circ - C) = A + 2C - 180^\circ$$

$$= A + 2C - (A + B + C) = C - B.$$

$$\text{Also } OA = R \text{ and } PA = 2R \cos A.$$

Now in $\triangle AOP$,

$$OP^2 = OA^2 + PA^2 - 2OA \cdot PA \cos \angle OAP$$



$$= R^2 + 4R^2 \cos^2 A - 4R^2 \cos A \cos(C - B)$$

$$= R^2 + 4R^2 \cos A [\cos A - \cos(C - B)]$$

$$= R^2 - 4R^2 \cos A [\cos(B + C) + \cos(C - B)] = R^2 - 8R^2 \cos A \cos B \cos C.$$

$$\text{Hence } OP = R\sqrt{1 - 8\cos A \cos B \cos C}.$$

Ans.

13. SOLUTION OF TRIANGLES :

The three sides a, b, c and the three angles A, B, C are called the elements of the triangle ABC . When any three of these six elements (except all the three angles) of a triangle are given, the triangle is known completely; that is the other three elements can be expressed in terms of the given elements and can be evaluated. This process is called the solution of triangles.

* If the three sides a, b, c are given, angle A is obtained from $\tan \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$

or $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$. B and C can be obtained in the similar way.

* If two sides b and c and the included angle A are given, then $\tan \frac{B-C}{2} = \frac{b-c}{b+c} \cot \frac{A}{2}$ gives

$\frac{B-C}{2}$. Also $\frac{B+C}{2} = 90^\circ - \frac{A}{2}$, so that B and C can be evaluated. The third side is given by

$$a = b \frac{\sin A}{\sin B}$$

or $a^2 = b^2 + c^2 - 2bc \cos A$.

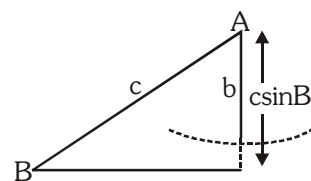
* If two sides b and c and an angle opposite the one of them (say B) are given then

$$\sin C = \frac{c}{b} \sin B, \quad A = 180^\circ - (B + C) \quad \text{and} \quad a = \frac{b \sin A}{\sin B} \quad \text{given the remaining elements.}$$

Case I :

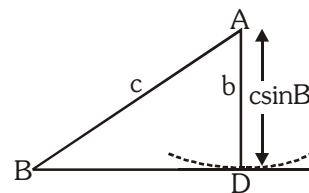
$$b < c \sin B.$$

We draw the side c and angle B . Now it is obvious from the figure that there is no triangle possible.

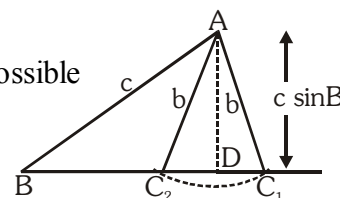


Case II :

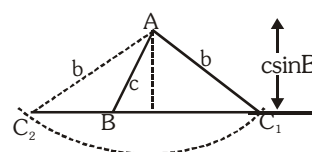
$b = c \sin B$ and B is an acute angle, there is only one triangle possible. and it is right-angled at C .

**Case III :**

$b > c \sin B$, $b < c$ and B is an acute angle, then there are two triangles possible for two values of angle C .

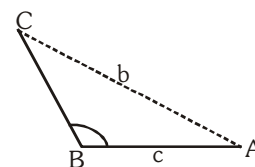
**Case IV :**

$b > c \sin B$, $c < b$ and B is an acute angle, then there is only one triangle.

**Case V :**

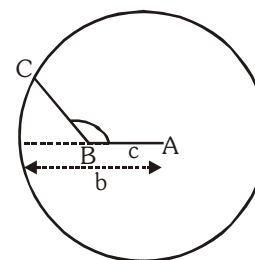
$b > c \sin B$, $c > b$ and B is an obtuse angle.

For any choice of point C , b will be greater than c which is a contradiction as $c > b$ (given). So there is no triangle possible.

**Case VI :**

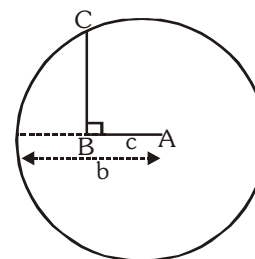
$b > c \sin B$, $c < b$ and B is an obtuse angle.

We can see that the circle with A as centre and b as radius will cut the line only in one point. So only one triangle is possible.

**Case VII :**

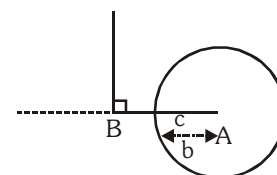
$b > c$ and $B = 90^\circ$.

Again the circle with A as centre and b as radius will cut the line only in one point. So only one triangle is possible.

**Case VIII :**

$b \leq c$ and $B = 90^\circ$.

The circle with A as centre and b as radius will not cut the line in any point. So no triangle is possible.



This is, sometimes, called an ambiguous case.

Alternative Method :

By applying cosine rule, we have $\cos B = \frac{a^2 + c^2 - b^2}{2ac}$

$$\Rightarrow a^2 - (2c \cos B)a + (c^2 - b^2) = 0 \Rightarrow a = c \cos B \pm \sqrt{(c \cos B)^2 - (c^2 - b^2)}$$

$$\Rightarrow a = c \cos B \pm \sqrt{b^2 - (c \sin B)^2}$$

This equation leads to following cases :

Case-I : If $b < c \sin B$, no such triangle is possible.

Case-II : Let $b = c \sin B$. There are further following case :

(a) B is an obtuse angle $\Rightarrow \cos B$ is negative. There exists no such triangle.

(b) B is an acute angle $\Rightarrow \cos B$ is positive. There exists only one such triangle.

Case-III : Let $b > c \sin B$. There are further following cases :

(a) B is an acute angle $\Rightarrow \cos B$ is positive. In this case triangle will exist if and only if

$c \cos B > \sqrt{b^2 - (c \sin B)^2}$ or $c > b \Rightarrow$ Two such triangle is possible. If $c < b$, only one such triangle is possible.

(b) B is an obtuse angle $\Rightarrow \cos B$ is negative. In this case triangle will exist if and only if

$\sqrt{b^2 - (c \sin B)^2} > |c \cos B| \Rightarrow b > c$. So in this case only one such triangle is possible. If

$b < c$ there exists no such triangle.

This is called an ambiguous case.

* If one side a and angles B and C are given, then $A = 180^\circ - (B + C)$, and $b = \frac{a \sin B}{\sin A}$, $c = \frac{a \sin C}{\sin A}$.

* If the three angles A,B,C are given, we can only find the ratios of the sides a,b,c by using sine rule (since there are infinite similar triangles possible).

Illustration 15 : In the ambiguous case of the solution of triangles, prove that the circumcircles of the two triangles are of same size.

Solution : Let us say b,c and angle B are given in the ambiguous case. Both the triangles will have b and its opposite angle as B. so $\frac{b}{\sin B} = 2R$ will be given for both the triangles. So their circumradii and therefore their sizes will be same.

Illustration 16 : If a, b and A are given in a triangle and c_1, c_2 are the possible values of the third side, prove that $c_1^2 + c_2^2 - 2c_1c_2 \cos 2A = 4a^2 \cos^2 A$.

Solution :

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\Rightarrow c^2 - 2bc \cos A + b^2 - a^2 = 0.$$

$$c_1 + c_2 = 2b \cos A \text{ and } c_1 c_2 = b^2 - a^2.$$

$$\Rightarrow c_1^2 + c_2^2 - 2c_1 c_2 \cos 2A = (c_1 + c_2)^2 - 2c_1 c_2 (1 + \cos 2A)$$

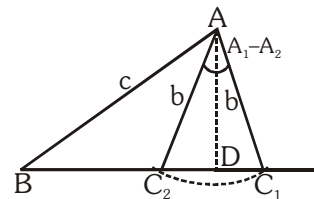
$$= 4b^2 \cos^2 A - 2(b^2 - a^2) 2 \cos^2 A = 4a^2 \cos^2 A.$$

Illustration 17 : If b, c, B are given and $b < c$, prove that $\cos\left(\frac{A_1 - A_2}{2}\right) = \frac{c \sin B}{b}$.

Solution : $\angle C_2 A C_1$ is bisected by AD .

$$\Rightarrow \text{In } \triangle A C_2 D, \cos\left(\frac{A_1 - A_2}{2}\right) = \frac{AD}{AC_2} = \frac{c \sin B}{b}$$

Hence proved.



Do yourself - 9 :

- If b, c, B are given and $b < c$, prove that $\sin\left(\frac{A_1 - A_2}{2}\right) = \frac{a_1 - a_2}{2b}$
- In a $\triangle ABC$, b, c, B ($c > b$) are given. If the third side has two values a_1 and a_2 such that

$$a_1 = 3a_2, \text{ show that } \sin B = \sqrt{\frac{4b^2 - c^2}{3c^2}}.$$

14. REGULAR POLYGON :

A regular polygon has all its sides equal. It may be inscribed or circumscribed.

(a) **Inscribed in circle of radius r :**

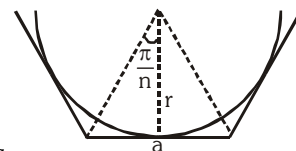
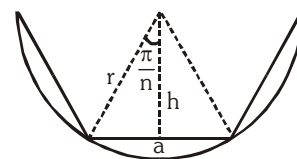
- $a = 2h \tan \frac{\pi}{n} = 2r \sin \frac{\pi}{n}$
- Perimeter (P) and area (A) of a regular polygon of n sides inscribed in a circle of radius r are given by $P = 2nr \sin \frac{\pi}{n}$ and $A = \frac{1}{2} nr^2 \sin \frac{2\pi}{n}$

(b) **Circumscribed about a circle of radius r :**

- $a = 2r \tan \frac{\pi}{n}$
- Perimeter (P) and area (A) of a regular polygon of n sides

circumscribed about a given circle of radius r is given by $P = 2nr \tan \frac{\pi}{n}$ and

$$A = nr^2 \tan \frac{\pi}{n}$$



Do yourself - 10 :

- (i) If the perimeter of a circle and a regular polygon of n sides are equal, then

prove that $\frac{\text{area of the circle}}{\text{area of polygon}} = \frac{\tan \frac{\pi}{n}}{\frac{\pi}{n}}$.

- (ii) The ratio of the area of n -sided regular polygon, circumscribed about a circle, to the area of the regular polygon of equal number of sides inscribed in the circle is $4 : 3$. Find the value of n .

15. SOME NOTES :

- (a) (i) If $a \cos B = b \cos A$, then the triangle is isosceles.
 (ii) If $a \cos A = b \cos B$, then the triangle is isosceles or right angled.
- (b) In right angle triangle
 (i) $a^2 + b^2 + c^2 = 8R^2$ (ii) $\cos^2 A + \cos^2 B + \cos^2 C = 1$
- (c) In equilateral triangle
 (i) $R = 2r$ (ii) $r_1 = r_2 = r_3 = \frac{3R}{2}$
 (iii) $r : R : r_1 = 1 : 2 : 3$ (iv) $\text{area} = \frac{\sqrt{3}a^2}{4}$ (v) $R = \frac{a}{\sqrt{3}}$
- (d) (i) The circumcentre lies (1) inside an acute angled triangle (2) outside an obtuse angled triangle & (3) mid point of the hypotenuse of right angled triangle.
 (ii) The orthocentre of right angled triangle is the vertex at the right angle.
 (iii) The orthocentre, centroid & circumcentre are collinear & centroid divides the line segment joining orthocentre & circumcentre internally in the ratio $2 : 1$ except in case of equilateral triangle. In equilateral triangle, all these centres coincide
- (e) Area of a cyclic quadrilateral $= \sqrt{(s-a)(s-b)(s-c)(s-d)}$

where a, b, c, d are lengths of the sides of quadrilateral and $s = \frac{a+b+c+d}{2}$.

ANSWERS FOR DO YOURSELF

- 1: (i) 90°
- 5: (i) (a) $\frac{3}{5}$ (b) $\frac{3}{4}$ (c) $\frac{1}{\sqrt{10}}$ (d) $\frac{3}{\sqrt{10}}$ (e) $\frac{1}{3}$ (f) 24
- 6: (i) (a) 6 (b) $\frac{5}{2}$ (c) 1
- 7: (i) (a) 1 (b) 3 (c) $2\sqrt{3}$
- 10: (ii) 6

ELEMENTARY EXERCISE

- Angles A, B and C of a triangle ABC are in A.P. If $\frac{b}{c} = \sqrt{\frac{3}{2}}$, then $\angle A$ is equal to
 (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{5\pi}{12}$ (D) $\frac{\pi}{2}$
- If K is a point on the side BC of an equilateral triangle ABC and if $\angle BAK = 15^\circ$, then the ratio of lengths $\frac{AK}{AB}$ is
 (A) $\frac{3\sqrt{2}(3+\sqrt{3})}{2}$ (B) $\frac{\sqrt{2}(3+\sqrt{3})}{2}$ (C) $\frac{\sqrt{2}(3-\sqrt{3})}{2}$ (D) $\frac{3\sqrt{2}(3-\sqrt{3})}{2}$
- In a triangle ABC, $\angle A = 60^\circ$ and $b : c = (\sqrt{3} + 1) : 2$ then $(\angle B - \angle C)$ has the value equal to
 (A) 15° (B) 30° (C) 22.5° (D) 45°
- In an acute triangle ABC, $\angle ABC = 45^\circ$, $AB = 3$ and $AC = \sqrt{6}$. The angle $\angle BAC$, is
 (A) 60° (B) 65° (C) 75° (D) 15° or 75°
- Let ABC be a right triangle with length of side $AB = 3$ and hypotenuse $AC = 5$.
 If D is a point on BC such that $\frac{BD}{DC} = \frac{AB}{AC}$, then AD is equal to
 (A) $\frac{4\sqrt{3}}{3}$ (B) $\frac{3\sqrt{5}}{2}$ (C) $\frac{4\sqrt{5}}{3}$ (D) $\frac{5\sqrt{3}}{4}$
- In a triangle ABC, if $a = 6$, $b = 3$ and $\cos(A - B) = \frac{4}{5}$, the area of the triangle is
 (A) 8 (B) 9 (C) 12 (D) $\frac{15}{2}$
- In $\triangle ABC$, if $a = 2b$ and $A = 3B$, then the value of $\frac{c}{b}$ is equal to
 (A) 3 (B) $\sqrt{2}$ (C) 1 (D) $\sqrt{3}$
- If the sides of a triangle are $\sin \alpha$, $\cos \alpha$, $\sqrt{1 + \sin \alpha \cos \alpha}$, $0 < \alpha < \frac{\pi}{2}$, the largest angle is
 (A) 60° (B) 90° (C) 120° (D) 150°
- If the angle A, B and C of a triangle are in an arithmetic progression and if a, b and c denote the lengths of the sides opposite to A, B and C respectively, then the value of expression

$$E = \left(\frac{a}{c} \sin 2C + \frac{c}{a} \sin 2A \right)$$
, is
 (A) $\frac{1}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) 1 (D) $\sqrt{3}$

10. If in a triangle $\sin A : \sin C = \sin(A - B) : \sin(B - C)$, then a^2, b^2, c^2
 (A) are in A.P. (B) are in G.P. (C) are in H.P. (D) none of these
11. In triangle ABC, if $\cot \frac{A}{2} = \frac{b+c}{a}$, then triangle ABC must be
 [Note: All symbols used have usual meaning in $\triangle ABC$.]
 (A) isosceles (B) equilateral (C) right angled (D) isosceles right angled
12. Consider a triangle ABC and let a, b and c denote the lengths of the sides opposite to vertices A, B and C respectively. If $a = 1, b = 3$ and $C = 60^\circ$, then $\sin^2 B$ is equal to
 (A) $\frac{27}{28}$ (B) $\frac{3}{28}$ (C) $\frac{81}{28}$ (D) $\frac{1}{3}$
13. The ratio of the sides of a triangle ABC is $1 : \sqrt{3} : 2$. Then ratio of $A : B : C$ is
 (A) $3 : 5 : 2$ (B) $1 : \sqrt{3} : 2$ (C) $3 : 2 : 1$ (D) $1 : 2 : 3$
14. In triangle ABC, If $s = 3 + \sqrt{3} + \sqrt{2}$, $3B - C = 30^\circ$, $A + 2B = 120^\circ$, then the length of longest side of triangle is
 [Note: All symbols used have usual meaning in triangle ABC.]
 (A) 2 (B) $2\sqrt{2}$ (C) $2(\sqrt{3} + 1)$ (D) $\sqrt{3} - 1$
15. In a triangle $\tan A : \tan B : \tan C = 1 : 2 : 3$, then $a^2 : b^2 : c^2$ equals
 (A) $5 : 8 : 9$ (B) $5 : 8 : 12$ (C) $3 : 5 : 8$ (D) $5 : 8 : 10$
16. In $\triangle ABC$, if a, b, c (taken in that order) are in A.P. then $\cot \frac{A}{2} \cot \frac{C}{2} =$
 [Note: All symbols used have usual meaning in triangle ABC.]
 (A) 1 (B) 2 (C) 3 (D) 4
17. In $\triangle ABC$ if $a = 8, b = 9, c = 10$, then the value of $\frac{\tan C}{\sin B}$ is
 (A) $\frac{32}{9}$ (B) $\frac{24}{7}$ (C) $\frac{21}{4}$ (D) $\frac{18}{5}$
18. In triangle ABC, if $\Delta = a^2 - (b - c)^2$, then $\tan A =$
 [Note: All symbols used have usual meaning in triangle ABC.]
 (A) $\frac{15}{16}$ (B) $\frac{1}{2}$ (C) $\frac{8}{17}$ (D) $\frac{8}{15}$
19. In a triangle ABC, if the sides a, b, c are roots of $x^3 - 11x^2 + 38x - 40 = 0$. If $\sum \left(\frac{\cos A}{a} \right) = \frac{p}{q}$, then find the least value of $(p + q)$ where $p, q \in \mathbb{N}$.
20. ABC is a triangle such that $\sin(2A + B) = \sin(C - A) = -\sin(B + 2C) = \frac{1}{2}$. If A, B, C are in A.P., find A, B, C.

EXERCISE (O-1)

1. A triangle has vertices A, B and C, and the respective opposite sides have lengths a, b and c. This triangle is inscribed in a circle of radius R. If $b = c = 1$ and the altitude from A to side BC has length $\sqrt{\frac{2}{3}}$, then R equals
- (A) $\frac{1}{\sqrt{3}}$ (B) $\frac{2}{\sqrt{3}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{\sqrt{3}}{2\sqrt{2}}$
2. A circle is inscribed in a right triangle ABC, right angled at C. The circle is tangent to the segment AB at D and length of segments AD and DB are 7 and 13 respectively. Area of triangle ABC is equal to
- (A) 91 (B) 96 (C) 100 (D) 104
3. In a triangle ABC, if $a = 13$, $b = 14$ and $c = 15$, then angle A is equal to
(All symbols used have their usual meaning in a triangle.)
- (A) $\sin^{-1} \frac{4}{5}$ (B) $\sin^{-1} \frac{3}{5}$ (C) $\sin^{-1} \frac{3}{4}$ (D) $\sin^{-1} \frac{2}{3}$
4. In a triangle ABC, if $b = (\sqrt{3} - 1)a$ and $\angle C = 30^\circ$, then the value of $(A - B)$ is equal to
(All symbols used have usual meaning in a triangle.)
- (A) 30° (B) 45° (C) 60° (D) 75°
5. In triangle ABC, if $AC = 8$, $BC = 7$ and D lies between A and B such that $AD = 2$, $BD = 4$, then the length CD equals
- (A) $\sqrt{46}$ (B) $\sqrt{48}$ (C) $\sqrt{51}$ (D) $\sqrt{75}$
6. In a triangle ABC, if $\angle C = 105^\circ$, $\angle B = 45^\circ$ and length of side AC = 2 units, then the length of the side AB is equal to
- (A) $\sqrt{2}$ (B) $\sqrt{3}$ (C) $\sqrt{2} + 1$ (D) $\sqrt{3} + 1$
7. In a triangle ABC, if $(a + b + c)(a + b - c)(b + c - a)(c + a - b) = \frac{8a^2b^2c^2}{a^2 + b^2 + c^2}$, then the triangle is
- [Note: All symbols used have usual meaning in triangle ABC.]
- (A) isosceles (B) right angled (C) equilateral (D) obtuse angled
8. In triangle ABC, if $2b = a + c$ and $A - C = 90^\circ$, then $\sin B$ equals
- [Note: All symbols used have usual meaning in triangle ABC.]
- (A) $\frac{\sqrt{7}}{5}$ (B) $\frac{\sqrt{5}}{8}$ (C) $\frac{\sqrt{7}}{4}$ (D) $\frac{\sqrt{5}}{3}$

9. In a triangle ABC, $a^3 + b^3 + c^3 = c^2(a + b + c)$

(All symbol used have usual meaning in a triangle.)

Statement-1: The value of $\angle C = 60^\circ$.

Statement -2: $\triangle ABC$ must be equilateral.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

10. The sides of a triangle are three consecutive integers. The largest angle is twice the smallest one.

The area of triangle is equal to

- (A) $\frac{5}{4}\sqrt{7}$ (B) $\frac{15}{2}\sqrt{7}$ (C) $\frac{15}{4}\sqrt{7}$ (D) $5\sqrt{7}$

11. The sides a, b, c (taken in that order) of triangle ABC are in A.P.

If $\cos \alpha = \frac{a}{b+c}$, $\cos \beta = \frac{b}{c+a}$, $\cos \gamma = \frac{c}{a+b}$ then $\tan^2\left(\frac{\alpha}{2}\right) + \tan^2\left(\frac{\gamma}{2}\right)$ is equal to

[Note: All symbols used have usual meaning in triangle ABC.]

- (A) 1 (B) $\frac{1}{2}$ (C) $\frac{1}{3}$ (D) $\frac{2}{3}$

12. AD and BE are the medians of a triangle ABC. If $AD = 4$, $\angle DAB = \frac{\pi}{6}$, $\angle ABE = \frac{\pi}{3}$, then area of triangle ABC equals

- (A) $\frac{8}{3}$ (B) $\frac{16}{3}$ (C) $\frac{32}{3}$ (D) $\frac{32}{9}\sqrt{3}$

13. In triangle ABC, if $\sin^3 A + \sin^3 B + \sin^3 C = 3 \sin A \cdot \sin B \cdot \sin C$, then triangle is

- (A) obtuse angled (B) right angled (C) obtuse right angled (D) equilateral

14. For right angled isosceles triangle, $\frac{r}{R} =$

[Note: All symbols used have usual meaning in triangle ABC.]

- (A) $\tan \frac{\pi}{12}$ (B) $\cot \frac{\pi}{12}$ (C) $\tan \frac{\pi}{8}$ (D) $\cot \frac{\pi}{8}$

15. In triangle ABC, If $\frac{1}{a+c} + \frac{1}{b+c} = \frac{3}{a+b+c}$ then angle C is equal to

[Note: All symbols used have usual meaning in triangle ABC.]

- (A) 30° (B) 45° (C) 60° (D) 90°

EXERCISE (O-2)

Multiple Correct Answer Type :

1. In a triangle ABC, let $2a^2 + 4b^2 + c^2 = 2a(2b + c)$, then which of the following holds good?

[Note: All symbols used have usual meaning in a triangle.]

(A) $\cos B = \frac{-7}{8}$

(B) $\sin(A - C) = 0$

(C) $\frac{r}{r_1} = \frac{1}{5}$

(D) $\sin A : \sin B : \sin C = 1 : 2 : 1$

2. In a triangle ABC, if $a = 4$, $b = 8$, $\angle C = 60^\circ$, then which of the following relations is (are) correct?

[Note: All symbols used have usual meaning in triangle ABC.]

(A) The area of triangle ABC is $8\sqrt{3}$

(B) The value of $\sum \sin^2 A = 2$

(C) Inradius of triangle ABC is $\frac{2\sqrt{3}}{3 + \sqrt{3}}$

(D) The length of internal angle bisector of angle C is $\frac{4}{\sqrt{3}}$

3. In which of the following situations, it is possible to have a triangle ABC?

(All symbols used have usual meaning in a triangle.)

(A) $(a + c - b)(a - c + b) = 4bc$

(B) $b^2 \sin 2C + c^2 \sin 2B = ab$

(C) $a = 3$, $b = 5$, $c = 7$ and $C = \frac{2\pi}{3}$

(D) $\cos\left(\frac{A - C}{2}\right) = \cos\left(\frac{A + C}{2}\right)$

4. In a triangle ABC, which of the following quantities denote the area of the triangle?

(A) $\frac{a^2 - b^2}{2} \left(\frac{\sin A \sin B}{\sin(A - B)} \right)$

(B) $\frac{r_1 r_2 r_3}{\sqrt{\sum r_1 r_2}}$

(C) $\frac{a^2 + b^2 + c^2}{\cot A + \cot B + \cot C}$

(D) $r^2 \cot \frac{A}{2} \cdot \cot \frac{B}{2} \cot \frac{C}{2}$

5. In $\triangle ABC$, angle A, B and C are in the ratio $1 : 2 : 3$, then which of the following is (are) correct?

(All symbol used have usual meaning in a triangle.)

(A) Circumradius of $\triangle ABC = c$

(B) $a : b : c = 1 : \sqrt{3} : 2$

(C) Perimeter of $\triangle ABC = 3 + \sqrt{3}$

(D) Area of $\triangle ABC = \frac{\sqrt{3}}{8} c^2$

6. Let one angle of a triangle be 60° , the area of triangle is $10\sqrt{3}$ and perimeter is 20 cm. If $a > b > c$ where a, b and c denote lengths of sides opposite to vertices A, B and C respectively, then which of the following is (are) correct?

(A) Inradius of triangle is $\sqrt{3}$

(B) Length of longest side of triangle is 7

(C) Circumradius of triangle is $\frac{7}{\sqrt{3}}$

(D) Radius of largest escribed circle is $\frac{1}{12}$

7. In triangle ABC, let $b = 10$, $c = 10\sqrt{2}$ and $R = 5\sqrt{2}$ then which of the following statement(s) is (are) correct?

[Note: All symbols used have usual meaning in triangle ABC.]

- (A) Area of triangle ABC is 50.
 (B) Distance between orthocentre and circumcentre is $5\sqrt{2}$
 (C) Sum of circumradius and inradius of triangle ABC is equal to 10
 (D) Length of internal angle bisector of $\angle ACB$ of triangle ABC is $\frac{5}{2\sqrt{2}}$
8. In a triangle ABC, let $BC = 1$, $AC = 2$ and measure of angle C is 30° . Which of the following statement(s) is (are) correct?
- (A) $2 \sin A = \sin B$
 (B) Length of side AB equals $5 - 2\sqrt{3}$
 (C) Measure of angle A is less than 30°
 (D) Circumradius of triangle ABC is equal to length of side AB
9. Given an acute triangle ABC such that $\sin C = \frac{4}{5}$, $\tan A = \frac{24}{7}$ and $AB = 50$. Then-

(A) centroid, orthocentre and incentre of $\triangle ABC$ are collinear

(B) $\sin B = \frac{4}{5}$

(C) $\sin B = \frac{4}{7}$

(D) area of $\triangle ABC = 1200$

10. In a triangle ABC, if $\cos A \cos 2B + \sin A \sin 2B \sin C = 1$, then

(A) A, B, C are in A.P. (B) B, A, C are in A.P. (C) $\frac{r}{R} = 2$ (D) $\frac{r}{R} = \sqrt{2} \sin \frac{\pi}{12}$

11. In $\triangle ABC$, angle A is 120° , $BC + CA = 20$ and $AB + BC = 21$, then

(A) $AB > AC$

(B) $AB < AC$

(C) $\triangle ABC$ is isosceles

(D) area of $\triangle ABC = 14\sqrt{3}$

12. In a triangle ABC, $\angle A = 30^\circ$, $b = 6$. Let CB_1 and CB_2 are least and greatest integral value of side a for which two triangles can be formed. It is also given angle B_1 is obtuse and angle B_2 is acute angle.

(All symbols used have usual meaning in a triangle.)

(A) $|CB_1 - CB_2| = 1$

(B) $CB_1 + CB_2 = 9$

(C) area of $\triangle B_1CB_2 = 6 + \frac{3}{2}\sqrt{7}$

(D) area of $\triangle AB_2C = 6 + \frac{9}{2}\sqrt{3}$

13. If the lengths of the medians AD, BE and CF of triangle ABC are 6, 8, 10 respectively, then-
 (A) AD & BE are perpendicular (B) BE and CF are perpendicular
 (C) area of $\triangle ABC = 32$ (D) area of $\triangle DEF = 8$
14. Let P be an interior point of $\triangle ABC$.
 Match the correct entries for the ratios of the Area of $\triangle PBC$: Area of $\triangle PCA$: Area of $\triangle PAB$ depending on the position of the point P w.r.t. $\triangle ABC$.

Column-I

- (A) If P is centroid (G)
 (B) If P is incentre (I)
 (C) If P is orthocentre (H)
 (D) If P is circumcentre

Column-II

- (P) $\tan A : \tan B : \tan C$
 (Q) $\sin 2A : \sin 2B : \sin 2C$
 (R) $\sin A : \sin B : \sin C$
 (S) $1 : 1 : 1$
 (T) $\cos A : \cos B : \cos C$

EXERCISE (S-1)

1. Given a triangle ABC with sides $a = 7$, $b = 8$ and $c = 5$. If the value of the expression $\left(\sum \sin A\right)\left(\sum \cot \frac{A}{2}\right)$ can be expressed in the form $\frac{p}{q}$ where $p, q \in \mathbb{N}$ and $\frac{p}{q}$ is in its lowest form find the value of $(p + q)$.
2. If two times the square of the diameter of the circumcircle of a triangle is equal to the sum of the squares of its sides then prove that the triangle is right angled.
3. In acute angled triangle ABC, a semicircle with radius r_a is constructed with its base on BC and tangent to the other two sides. r_b and r_c are defined similarly. If r is the radius of the incircle of triangle ABC then prove that, $\frac{2}{r} = \frac{1}{r_a} + \frac{1}{r_b} + \frac{1}{r_c}$.
4. If the length of the perpendiculars from the vertices of a triangle A, B, C on the opposite sides are p_1, p_2, p_3 then prove that $\frac{1}{p_1} + \frac{1}{p_2} + \frac{1}{p_3} = \frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}$.
5. With usual notations, prove that in a triangle ABC
 $a \cot A + b \cot B + c \cot C = 2(R + r)$
6. With usual notations, prove that in a triangle ABC
 $Rr (\sin A + \sin B + \sin C) = \Delta$
7. With usual notations, prove that in a triangle ABC
 $\cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2} = \frac{s^2}{\Delta}$
8. With usual notations, prove that in a triangle ABC
 $\cot A + \cot B + \cot C = \frac{a^2 + b^2 + c^2}{4\Delta}$

9. If a, b, c are the sides of triangle ABC satisfying $\log\left(1 + \frac{c}{a}\right) + \log a - \log b = \log 2$.

Also $a(1 - x^2) + 2bx + c(1 + x^2) = 0$ has two equal roots. Find the value of $\sin A + \sin B + \sin C$.

10. With usual notations, prove that in a triangle ABC

$$\frac{b-c}{r_1} + \frac{c-a}{r_2} + \frac{a-b}{r_3} = 0$$

11. With usual notations, prove that in a triangle ABC

$$\frac{r_1}{(s-b)(s-c)} + \frac{r_2}{(s-c)(s-a)} + \frac{r_3}{(s-a)(s-b)} = \frac{3}{r}$$

12. With usual notations, prove that in a triangle ABC

$$\frac{abc}{s} \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} = \Delta$$

13. With usual notations, prove that in a triangle ABC

$$\frac{1}{r^2} + \frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} = \frac{a^2 + b^2 + c^2}{\Delta^2}$$

14. With usual notations, prove that in a triangle ABC

$$2R \cos A = 2R + r - r_1$$

15. If $r_1 = r + r_2 + r_3$ then prove that the triangle is a right angled triangle.

EXERCISE (S-2)

- With usual notation, if in a ΔABC , $\frac{b+c}{11} = \frac{c+a}{12} = \frac{a+b}{13}$; then prove that, $\frac{\cos A}{7} = \frac{\cos B}{19} = \frac{\cos C}{25}$.
- Given a triangle ABC with $AB = 2$ and $AC = 1$. Internal bisector of $\angle BAC$ intersects BC at D . If $AD = BD$ and Δ is the area of triangle ABC , then find the value of $12\Delta^2$.
- For any triangle ABC , if $B = 3C$, show that $\cos C = \sqrt{\frac{b+c}{4c}}$ & $\sin \frac{A}{2} = \frac{b-c}{2c}$.
- In a triangle ABC if $a^2 + b^2 = 101c^2$ then find the value of $\frac{\cot C}{\cot A + \cot B}$.
- The two adjacent sides of a cyclic quadrilateral are 2 & 5 and the angle between them is 60° . If the area of the quadrilateral is $4\sqrt{3}$, find the remaining two sides.
- If in a ΔABC , $a = 6$, $b = 3$ and $\cos(A - B) = 4/5$ then find its area.
- In a ΔABC , (i) $\frac{a}{\cos A} = \frac{b}{\cos B}$ (ii) $2 \sin A \cos B = \sin C$
(iii) $\tan^2 \frac{A}{2} + 2 \tan \frac{A}{2} \tan \frac{C}{2} - 1 = 0$, prove that (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (i).
- Two sides of a triangle are of lengths $\sqrt{6}$ and 4 and the angle opposite to smaller side is 30° . How many such triangles are possible? Find the length of their third side and area.

9. The triangle ABC (with side lengths a, b, c as usual) satisfies $\log a^2 = \log b^2 + \log c^2 - \log (2bc \cos A)$. What can you say about this triangle?
10. The sides of a triangle are consecutive integers $n, n+1$ and $n+2$ and the largest angle is twice the smallest angle. Find n .

EXERCISE (JA)

1. Let ABC and ABC' be two non-congruent triangles with sides $AB = 4, AC = AC' = 2\sqrt{2}$ and angle $B = 30^\circ$. The absolute value of the difference between the areas of these triangles is [JEE 2009, 5]
2. (a) If the angle A, B and C of a triangle are in an arithmetic progression and if a, b and c denote the length of the sides opposite to A, B and C respectively, then the value of the expression

$$\frac{a}{c} \sin 2C + \frac{c}{a} \sin 2A, \text{ is -}$$

- (A) $\frac{1}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) 1 (D) $\sqrt{3}$

- (b) Consider a triangle ABC and let a, b and c denote the length of the sides opposite to vertices A, B and C respectively. Suppose $a = 6, b = 10$ and the area of the triangle is $15\sqrt{3}$. If $\angle ACB$ is obtuse and if r denotes the radius of the incircle of the triangle, then r^2 is equal to

- (c) Let ABC be a triangle such that $\angle ACB = \frac{\pi}{6}$ and let a, b and c denote the lengths of the sides opposite to A, B and C respectively. The value(s) of x for which $a = x^2 + x + 1, b = x^2 - 1$ and $c = 2x + 1$ is/are [JEE 2010, 3+3+3]

- (A) $-(2 + \sqrt{3})$ (B) $1 + \sqrt{3}$ (C) $2 + \sqrt{3}$ (D) $4\sqrt{3}$

3. Let PQR be a triangle of area Δ with $a = 2, b = \frac{7}{2}$ and $c = \frac{5}{2}$, where a, b and c are the lengths of the sides of the triangle opposite to the angles at P, Q and R respectively. Then $\frac{2 \sin P - \sin 2P}{2 \sin P + \sin 2P}$ equals [JEE 2012, 3M, -1M]

- (A) $\frac{3}{4\Delta}$ (B) $\frac{45}{4\Delta}$ (C) $\left(\frac{3}{4\Delta}\right)^2$ (D) $\left(\frac{45}{4\Delta}\right)^2$

4. In a triangle PQR, P is the largest angle and $\cos P = \frac{1}{3}$. Further the incircle of the triangle touches the sides PQ, QR and RP at N, L and M respectively, such that the lengths of PN, QL and RM are consecutive even integers. Then possible length(s) of the side(s) of the triangle is (are)

[JEE(Advanced) 2013, 3, (-1)]

- (A) 16 (B) 18 (C) 24 (D) 22

5. In a triangle the sum of two sides is x and the product of the same two sides is y . If $x^2 - c^2 = y$, where c is a third side of the triangle, then the ratio of the in-radius to the circum-radius of the triangle is -
[JEE(Advanced)-2014, 3(-1)]

(A) $\frac{3y}{2x(x+c)}$ (B) $\frac{3y}{2c(x+c)}$ (C) $\frac{3y}{4x(x+c)}$ (D) $\frac{3y}{4c(x+c)}$

6. In a triangle XYZ, let x, y, z be the lengths of sides opposite to the angles X, Y, Z , respectively and

$2s = x + y + z$. If $\frac{s-x}{4} = \frac{s-y}{3} = \frac{s-z}{2}$ and area of incircle of the triangle XYZ is $\frac{8\pi}{3}$, then-

(A) area of the triangle XYZ is $6\sqrt{6}$

(B) the radius of circumcircle of the triangle XYZ is $\frac{35}{6}\sqrt{6}$

(C) $\sin \frac{X}{2} \sin \frac{Y}{2} \sin \frac{Z}{2} = \frac{4}{35}$

(D) $\sin^2 \left(\frac{X+Y}{2} \right) = \frac{3}{5}$

[JEE(Advanced)-2016, 4(-2)]

7. In a triangle PQR, let $\angle PQR = 30^\circ$ and the sides PQ and QR have lengths $10\sqrt{3}$ and 10, respectively.

Then, which of the following statement(s) is (are) TRUE ? [JEE(Advanced)-2018, 4(-2)]

(A) $\angle QPR = 45^\circ$

(B) The area of the triangle PQR is $25\sqrt{3}$ and $\angle QRP = 120^\circ$

(C) The radius of the incircle of the triangle PQR is $10\sqrt{3} - 15$

(D) The area of the circumcircle of the triangle PQR is 100π .

ANSWERS

ELEMENTARY EXERCISE

1. C 2. C 3. B 4. C 5. B 6. B 7. D 8. C 9. D
 10. A 11. C 12. A 13. D 14. C 15. A 16. C 17. A 18. D
 19. 25 20. $45^\circ, 60^\circ, 75^\circ$

EXERCISE (O-1)

1. D 2. A 3. A 4. C 5. C 6. D 7. B 8. C
 9. C 10. C 11. D 12. D 13. D 14. C 15. C

EXERCISE (O-2)

1. B,C 2. A,B 3. B,C 4. A,B,D 5. B,D 6. A,C 7. A,B,C
 8. A,C,D 9. A,B,D 10. B,D 11. A,D 12. A,B,C,D 13. A,C,D
 14. (A) S; (B) R; (C) P; (D) Q

EXERCISE (S-1)

1. 107 9. $\frac{12}{5}$

EXERCISE (S-2)

2. 9 4. 50 5. 3 cms & 2 cms 6. 9 sq. unit
 8. Two triangle $(2\sqrt{3}-\sqrt{2})$, $(2\sqrt{3}+\sqrt{2})$, $(2\sqrt{3}-\sqrt{2})$ & $(2\sqrt{3}+\sqrt{2})$ sq. units
 9. triangle is isosceles 10. 4

EXERCISE (JA)

1. 4 2. (a) D, (b) 3, (c) B 3. C 4. B,D 5. B 6. A,C,D
 7. B,C,D